

PSMFlows: ZERO-SHOT REINFORCEMENT LEARNING ON LIMITED-SUPPORT DATA VIA FLOW-INDEXED SUCCESSOR MEASURES

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ABSTRACT

Zero-shot reinforcement learning methods such as Forward–Backward representations (FB) and Proto Successor Measures (PSM) learn task-agnostic occupancy representations from reward-free data, but their training loops evaluate candidate actions that the dataset never supports: value estimates at out-of-distribution actions are unconstrained, and the policy-improvement argmax seeks out exactly those errors. We propose *PSMFlows*, which removes this failure mode by construction. A behavior-cloned conditional flow $G_\theta(s, u)$ maps latent noise to data-like actions; holding the noise fixed defines a family of policies $\pi_u(s) = G_\theta(s, u)$ indexed by u , and a successor-measure representation $\psi(s, u', u)^\top \varphi(x)$ is learned over this family, with dataset transitions carrying their own latents via flow inversion. Every action the method considers—during training and at inference—is a decode of the frozen flow, converting the hard constraint $a \in \text{supp } \mu_{\mathcal{D}}(\cdot | s)$ into the benign constraint that $\|u\|$ be typical. We describe the three-stage pipeline, a staged diagnostic protocol that makes each assumption falsifiable in isolation, and report first results on OGBench: the flow, inversion, and latent-family stages certify cleanly, while first representation-learning runs isolate a concrete failure mode—the learned measure discriminates policy indices without aligning them to realized behavior—that our diagnostics localize and that motivates the next iteration of the training loss.

1 INTRODUCTION

Behavior foundation models promise a single reward-free pretraining phase after which any downstream reward can be solved zero-shot. The dominant family—FB (Touati & Ollivier, 2021) and PSM (Agarwal et al., 2024)—learns linear-algebraic representations of successor measures. Applied to *offline* data with limited support, however, these methods inherit the classic offline-RL pathology: their Bellman maximization queries actions the dataset cannot certify. Conservatism penalties (VC-FB; Jeon et al., 2024) treat the symptom in the loss; we instead change the policy space so the symptom cannot arise.

Contributions. (i) a formulation of zero-shot offline RL as successor-measure learning over a *flow-indexed* policy family, in which the training distribution has behavior-policy concentrability $C = 1$; (ii) an in-sample training algorithm in which dataset transitions carry their own latents via flow inversion, with a relaxed-posterior (Laplace + importance-sampled EM) preimage operator; (iii) a staged, falsifiable diagnostic protocol (D1–D4) that separates flow fidelity, index coherence, inversion health, and control; (iv) an empirical study on OGBench, including a negative result whose diagnosis is, we argue, itself informative about what successor-measure losses ground and what they do not.

2 PRELIMINARIES

Setting. A reward-free MDP $(\mathcal{S}, \mathcal{A}, P, \gamma)$ with $\mathcal{A} \subseteq \mathbb{R}^{d_a}$; an offline dataset $\mathcal{D} = \{(s_i, a_i, s'_i)\}_{i=1}^N$ collected by an unknown behavior policy $\mu(\cdot | s)$ with state marginal ρ . At test time a reward r is revealed (as labels on a sample of $\mathcal{D}$) and a policy must be produced with no further interaction.

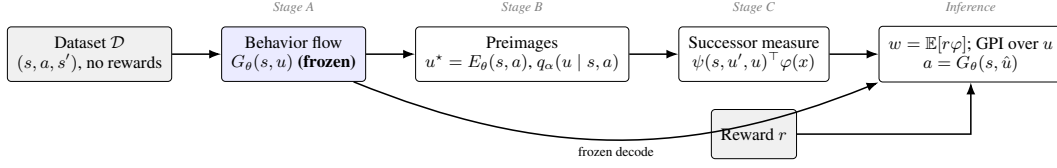


Figure 1: **The PSMFlows pipeline.** A reward-free behavior flow (Stage A, frozen) turns noise into data-like actions; flow inversion (Stage B) equips every dataset transition with its latent; a successor measure over the flow-indexed policy family is trained in-sample (Stage C); at test time a reward is projected onto the state basis and generalized policy improvement runs entirely in latent space. An editable draw.io source of this figure is provided in `figures/pipeline.drawio`.

Successor measures. For a policy π , $M^\pi(s, a, X) = \mathbb{E}[\sum_{t \geq 1} \gamma^{t-1} \mathbf{1}\{s_t \in X\} \mid s_0 = s, a_0 = a]$ is the discounted occupancy of $X \subseteq \mathcal{S}$. Then $Q_r^\pi(s, a) = \int r dM^\pi(s, a, \cdot)$: one object serves every reward, which is the basis of zero-shot RL. FB parameterizes $m^\pi(s, a, x) = F(s, a, z_\pi)^\top B(x)$; PSM uses an affine decomposition $m^\pi = b + \Phi w_\pi$ whose policy dependence is linear. Both train by a measure-matching temporal-difference loss and infer tasks by projecting r on the state basis.

The limited-support problem. The TD loss and the actor of these methods evaluate Q at actions from a training policy π_{tr} . The relevant quantity is the concentrability $C(\pi_{\text{tr}}) = \text{ess sup } \pi_{\text{tr}}(a \mid s) / \mu(a \mid s)$: FB’s deterministic actor gives $C = \infty$; PSM’s uniform action sampling gives C finite only for full-support behavior. On play or mode-restricted data neither holds, and value errors at unsupported actions are exactly the ones the argmax selects.

Conditional flow matching. A velocity field $v_\theta(s, u_t, t)$ trained with the CFM regression $\mathbb{E}\|v_\theta(s, (1-t)u_0 + ta, t) - (a - u_0)\|^2$, $u_0 \sim \mathcal{N}(0, I)$, induces a decoder $G_\theta(s, \cdot)$: the time-one map of the ODE $\dot{u} = v_\theta(s, u, t)$. With latent dimension equal to action dimension, $G_\theta(s, \cdot)$ is a C^1 diffeomorphism of $\mathbb{R}^{d_a}$; its inverse $E_\theta(s, \cdot)$ is computed by backward integration.

3 METHOD: PSMFLOWS

3.1 A FLOW-INDEXED POLICY FAMILY

Fix the trained (and thereafter frozen) behavior flow. For $u \in \mathbb{R}^{d_a}$ define the fixed-index policy $\pi_u(s) = G_\theta(s, u)$. Two properties make this family the right search space for offline zero-shot RL. First, *mass conversion*: for any set U of prior mass $1 - \delta$, the decoded actions carry $1 - \delta$ of the behavior mass *simultaneously at every state*, so “stay on the data” becomes “keep $\|u\|$ typical”. Second, *divergence isometry*: for any ν over latents, $D_f(\pi_\nu \parallel \hat{\mu}_\theta) = D_f(\nu \parallel p_0)$ at every state, so trust-region constraints against the (unknown) behavior density become closed-form constraints against a standard normal.

3.2 DATASET LATENTS BY FLOW INVERSION

Training a measure over $\{\pi_u\}$ requires transitions labeled with latents. Since $G_\theta(s, \cdot)$ is invertible, each dataset action has a point preimage $u_i^* = E_\theta(s_i, a_i)$; because a whole basin of latents decodes near a_i , we also fit the relaxed posterior

$$q_\alpha(u \mid s, a) \propto \mathcal{N}(u; 0, I) \exp(-\alpha \|G_\theta(s, u) - a\|), \quad (1)$$

by importance-sampled EM initialized from the Laplace approximation with covariance $\propto (J^\top J)^{-1}$, $J = \partial G_\theta / \partial u$ (Figure 2). The temperature α trades sharpness against effective sample size (ESS); both are measured and gated (§4). Under an exact flow, $E_\theta(s, a) \sim \mathcal{N}(0, I)$ on data, so the $\chi_{d_a}^2$ typicality of $\|u^*\|^2$ is a direct, falsifiable test of the flow-equals-behavior assumption.

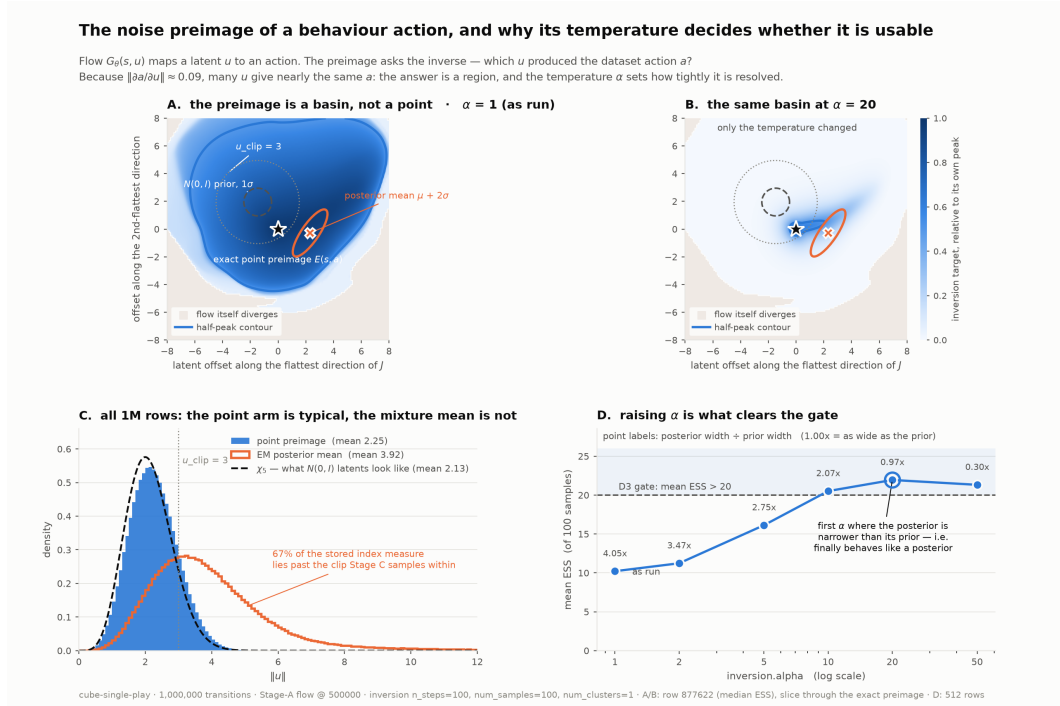


Figure 2: **What the noise preimage is, and why the temperature α decides whether it is usable.** (A) a 2D latent-space slice at one real cube transition: the preimage of an action is a *basin*, not a point; at $\alpha=1$ the inversion target (shading) is far broader than the $\mathcal{N}(0, I)$ prior (dashed). (B) the same slice at $\alpha=20$: the basin contracts to prior scale. (C) aggregate over 10^6 transitions: point preimages track the χ_{d_a} typical radius, while low- α posterior means drift far outside it. (D) mean ESS versus α against the D3 gate.

3.3 SUCCESSOR MEASURES OVER THE FAMILY

We learn $m(s, u, u', x) = \psi(s, u', u)^\top \varphi(x)$, the density of $M^{u \rightarrow u'}(s, \cdot)$: take the action indexed by u now, then follow $\pi_{u'}$. The Bellman identity closes on the diagonal, $M^{u \rightarrow u'}(s, X) = \mathbb{E}_{s'}[\mathbf{1}\{s' \in X\} + \gamma M^{u' \rightarrow u'}(s', X)]$, so a single network suffices. The empirical loss is the standard measure-matching TD objective with batch negatives, an orthonormality regularizer on φ , and three sampling rules: the action slot u is drawn from the stored preimage posterior of the transition (clipped to the typical set); the index slot u' is drawn fresh (mixed $\mathcal{N}(0, I)$ /permuted behavior latents); the bootstrap target is $\bar{\psi}(s', u', u')$ —the continuation latent *is* the policy index. No decoded action appears anywhere in training; under an exact flow the training distribution has concentrability $C = 1$ w.r.t. the behavior policy.

3.4 INFERENCE

Given rewards on relabeled data, $w = \frac{1}{N} \sum_i r(s'_i) \varphi(s'_i)$ is the least-squares task projection (exact under orthonormal φ), and $\psi(s, u', u)^\top w$ estimates the corresponding Q . Rung 1 (used here) is sampled GPI: draw K candidate latents and M indices, take $\hat{u} = \arg \max_k \max_m \psi(s, u'_m, u_k)^\top w$, act $a = G_\theta(s, \hat{u})$. Rungs 2–3 (in-sample latent Q -iteration; an amortized latent actor trained with a flow-BC anchor in latent space) are implemented but not yet part of the evaluated protocol.

4 EXPERIMENTS AND CURRENT FINDINGS

Protocol. OGBench (Park et al., 2024) `pointmaze-medium-navigate` and `cube-single-play` (state-based, single-task rewards); FQL (Park et al., 2025) as the per-task topline; in-repo, parity-verified PSM and FB as zero-shot baselines. All reported numbers use ≥ 3 seeds, 50-episode evaluations on a 50k-step grid, mean $\pm$ 95% CI; gates D1–D4 must pass in order, which localizes any failure to a single stage.

Gate	Question	Result (pointmaze / cube)	Status
D1	flow reproduces data actions	MMD $3 \cdot 10^{-4} / 1.5 \cdot 10^{-4}$ (ctrl. $\times 13 / \times 10^3$)	pass
D2	fixed u = distinct, repeatable policy	consistency ratio 0.53 / 0.51	pass
D3	inversion trustworthy	ESS 91 / 82; roundtrip $\sim 10^{-4}$; typicality ≥ 0.99	pass
D4	oracle-reward control $\geq 50\%$ of FQL	0.00 vs. FQL peak 0.687 ± 0.201	fail

Table 1: **Staged diagnostics (current state).** The generative substrate certifies cleanly; control does not yet. The point of the protocol is that the failure is localized: it lies in the representation/inference stage, not the flow or the inversion.

Finding 1: the substrate works. The behavior flow beats random controls by 1–3 orders of magnitude on distribution match (D1); fixed-latent policies are reproducible and mutually distinct (D2); and at $\alpha=20$, $N=200$ the preimage posterior passes every inversion gate (D3), with 99% of inverted latents inside the central χ_{da}^2 typicality band—direct evidence for the exact-flow assumption at the fidelity our gates demand.

Finding 2: the temperature is measurable, and it matters. At $\alpha=1$ the Laplace proposal collapses to the prior (99.8% of rows fully clipped), 39% of posterior means leave the typical set, and ESS ≈ 10 ; at $\alpha=20$ posterior means track the χ_{da} radius and ESS exceeds the gate five-fold (Figure 2). Latents computed at the wrong temperature are quantifiably noise.

Finding 3 (negative, diagnosed): the measure discriminates indices it does not ground. First representation runs (3 seeds, 500k–1M steps, both preimage modes) reach 0.00 success even with oracle rewards. Gradient probing of the trained $\psi^\top w$ shows sensitivity 386 to the index slot u' and 115 to the state, but only 12.6 to the action slot u that GPI selects and decodes; the value landscape over candidate latents is flat to $\pm 2.5\%$, and predicted scores are uncorrelated with the realized returns of the corresponding fixed- u policies (Spearman -0.13). Meanwhile the *state*-value $V(s) = \max_u \psi(s, u, u)^\top w$ correlates 0.89 with negative distance-to-goal: the representation learns *where* the task is, but not *which policy index* realizes it—TD grounds the index slot only through bootstrap chains, never against the actual rollouts of $\pi_{u'}$.

5 RELATED WORK

Deferred to the next draft; see RESEARCH_NOTE §6 for the verified map.

6 CONCLUSION

PSMFlows converts the support constraint of offline zero-shot RL into latent-space typicality by construction. Its generative substrate is validated; its open problem—grounding the policy-index axis of the successor measure against realized behavior—is now precisely characterized, and is the subject of ongoing work.

REPRODUCIBILITY STATEMENT

All gates, tools, and launchers are scripted; every diagnostic writes a JSON artifact; preimage archives carry sidecars recording the exact checkpoint and inversion settings that produced them.

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A APPENDIX